# ELEC 334 - Microprocessors

## Midterm 2021 - 94fd35fa19187615cf582bafd190371e

name lastname: _____ student id: _____

| | | | |
|---|---|---|---|
| 1 | A B C D E | 21 | A B C D E |
| 2 | A B C D E | 22 | A B C D E |
| 3 | A B C D E | 23 | A B C D E |
| 4 | A B C D E | 24 | A B C D E |
| 5 | A B C D E | 25 | A B C D E |
| 6 | A B C D E | 26 | A B C D E |
| 7 | A B C D E | 27 | A B C D E |
| 8 | A B C D E | 28 | A B C D E |
| 9 | A B C D E | 29 | A B C D E |
| 10 | A B C D E | 30 | A B C D E |
| 11 | A B C D E | 31 | A B C D E |
| 12 | A B C D E | 32 | A B C D E |
| 13 | A B C D E | 33 | A B C D E |
| 14 | A B C D E | 34 | A B C D E |
| 15 | A B C D E | 35 | A B C D E |
| 16 | A B C D E | 36 | A B C D E |
| 17 | A B C D E | 37 | A B C D E |
| 18 | A B C D E | 38 | A B C D E |
| 19 | A B C D E | 39 | A B C D E |
| 20 | A B C D E | 40 | A B C D E |

**Question 1** - Which of the following *most accurately* describes the use of the `static` keyword?

A. A variable declared static within the body of a function has no meaning.
B. A function declared static cannot be called and will be optimized.
C. A variable declared static within the body of a function maintains its value between function invocations.
D. A variable declared static outside the body of a function has no meaning.
E. A variable declared static outside the body of a function can be accessed by any function.

**Question 2** - Let's assume `mem[0x1234] = 2222` and `mem[0x1238] = 4444` What is the content of `R0` after the following code?

```
1  LDR R0, =0x1234
2  LDR R1, =0x1238
3  LDR R0, [R0]
4  ADDS R0, R1
5
```

A. `0x4444`
B. `0x6666`
C. `0x3460`
D. `0x3450`
E. `0x2222`

**Question 3** - Which of the following is used to describe machines that use single complex instructions that can perform many operations?

A. KISC
B. von Neumann
C. FISC

D. LISC
E. CISC

**Question 4** - Which of the following best describes POP operation?

A. Memory content for the top of the stack is written to the register in the instruction, then pointing location is decremented by 4.
B. Memory content for the bottom of the stack is written to the register in the instruction, however, pointing location is not changed.
C. Memory content for the bottom of the stack is written to the register in the instruction, then pointing location is decremented by 4.
D. Memory content for the bottom of the stack is written to the register in the instruction, then pointing location is incremented by 4.
E. Memory content for the top of the stack is written to the register in the instruction, then pointing location is incremented by 4.

**Question 5** - Which of the following is used to describe an architecture that use a common bus for both data and code?

A. LISC
B. Harvard
C. Register-Register
D. von Neumann
E. FISC

**Question 6** - What is the `v[0]` value at the end of the following code for `v = [15, 25]` and `k = [1, 2, 3, 4]`? (There is one more question that has the same code snipped that asks for `v[1]`)

```
void big_problem (uint32_t* v, uint32_t* k) {
    uint32_t v0=v[0], v1=v[1], sum=0xC6EF3720;
    uint32_t delta=0x9e3779b9;
    uint32_t k0=k[0], k1=k[1], k2=k[2], k3=k[3];
    for (int i=0; i<1; i++) {
        v1 -= ((v0<<4) + k2) ^ (v0 + sum) ^ ((v0>>5) + k3);
        sum -= delta;
        v0 -= ((v1<<4) + k0) ^ (v1 + sum) ^ ((v1>>5) + k1);
    }
    v[0] = v0; v[1] = v1;
}

```

A. `0x98dc1452`
B. `0x47a23914`
C. `0xfbf33890`
D. `0x0ef37812`
E. `0x3910c841`

**Question 7** - Which of the following is used to describe an architecture that use separate buses for data and code?

A. von Neumann
B. LISC
C. Load-Store
D. FISC
E. Harvard

**Question 8** - What is the purpose of Program Status Register?

A. Hold information about program execution, exception number and ALU flags.
B. Keep the return address of a subroutine or function call.
C. Blocks interrupts.
D. Point to the top of the stack.
E. Point to the next instruction in line.

**Question 9** - Which of the following is not inside a program image?

A. heap code
B. vector table
C. startup code
D. reset handler
E. application code

**Question 10** - What is the race condition?

A. Unexpected result based on the timing of read and write operations.
B. Multiple definitions of the same variable.
C. Multiple weakly defiend function blocks.
D. Conflicting function signatures.
E. Order of the parameters getting mixed up when calling a function.

**Question 11** - What is the `v[1]` value at the end of the following code for `v = [15, 25]` and `k = [1, 2, 3, 4]`? (There is one more question that has the same code snipped that asks for `v[0]`)

```
void big_problem (uint32_t* v, uint32_t* k) {
    uint32_t v0=v[0], v1=v[1], sum=0xC6EF3720;
    uint32_t delta=0x9e3779b9;
    uint32_t k0=k[0], k1=k[1], k2=k[2], k3=k[3];
    for (int i=0; i<1; i++) {
        v1 -= ((v0<<4) + k2) ^ (v0 + sum) ^ ((v0>>5) + k3);
```

```
        sum -= delta;
        v0 -= ((v1<<4) + k0) ^ (v1 + sum) ^ ((v1>>5) + k1);
    }
    v[0] = v0; v[1] = v1;
}
```

A. `0x3910c841`
B. `0xfbf33890`
C. `0x0ef37812`
D. `0x98dc1452`
E. `0x47a23914`

**Question 12** - Which of the following is the *most portable way* to declare a C preprocessor constant for the number of seconds in a non-leap calendar year?

A. `#define SECONDS_PER_YEAR (60 * 60 * 24 * 365)`
B. `#define SECONDS_PER_YEAR (60 * 60 * 24) * 365`
C. `#define SECONDS_PER_YEAR 60 * 60 * 24 * 365`
D. `#define SECONDS_PER_YEAR 60 * 60 * 24 * 365;`
E. `#define SECONDS_PER_YEAR (60 * 60 * 24 * 365UL)`

**Question 13** - Which of the following is a memory usage mechanism that allows users to dynamically allocate space?

A. Heap
B. Stack
C. Book
D. Scratchpad
E. Page

**Question 14** - What are the first two addresses of the Vector Table?

A. Stack Pointer, Link Register
B. Program Counter, Link Register
C. Stack Pointer, Reset Vector
D. Link Register, Reset Vector
E. Program Counter, Stack Pointer

**Question 15** - Which of the following is a correct way to write the value `0xAA33` to the physical memory address `0x1234`?

A. `uint16_t* p= (uint16_t *) 0x1234; p = 0xAA33;`
B. `uint16_t p= (uint16_t) 0x1234; p = 0xAA33;`
C. `*(uint16_t * const)(0x1234) = 0xAA33;`
D. `uint16_t* p= (uint16_t *) 0xAA33; p = 0x1234;`
E. `*(uint16_t * const)(0xAA33) = 0x1234;`

**Question 16** - Which of the following is not a valid exception state?

A. Pending
B. Active
C. Active and Pending
D. Inactive
E. Suspended

**Question 17** - Which of the following can be defined as the programmer's view of the computer?

A. Instruction Set Architecture
B. Microcontroller
C. Processor
D. Pipeline
E. Microarchitecture

**Question 18** - Which of the following code snippets will reset the least-significant bit of x?

A. `x |= ~0x01;`
B. `x |= 0x01;`
C. `x | 0x01;`
D. `x &= ~0x01;`
E. `x &= 0x01;`

**Question 19** - Which of the following is a memory usage mechanism that allows the system memory to be used as temporary data storage?

A. Stack
B. Scratchpad
C. Page
D. Book
E. Heap

**Question 20** - Which of the following is a memory format that holds the lowest byte in lower address?

A. little endian
B. read write memory
C. read only memory
D. big endian
E. random access memory

**Question 21** - What is starving?

A. Functions (or tasks) not getting compiled at compile-time due to size optimization.
B. Functions (or tasks) not getting compatible parameters on run-time.
C. Functions (or tasks) not having enough time to run.
D. Functions (or tasks) not getting the necessary parameters on run-time.
E. Functions (or tasks) not getting compiled at compile-time due to speed optimization.

**Question 22** - What is the result of the following operation assuming `A = 0x1234`?

```
A &= ~0xFFDF;

```

A. `0xFFFF1234`
B. `0x00001234`
C. `0x1234FFDF`
D. `0x12340000`
E. `0x00000020`

**Question 23** - Which of the following is the most flexible way to declare a C preprocessor macro that takes two arhuments and returns the smaller of their two values?

A. `#define MIN(A, B) A < B ? A : B`
B. `#define MIN(A, B) do { A < B ? A : B} while(0)`
C. `#define MIN(A, B) (((A) < (B)) ? (A) : (B))`
D. `#define MIN(A, B) ((A < B) ? A : B)`
E. `#define MIN(A, B) { if (A < B) A; else B;}`

**Question 24** - Which of the following is usually used to describe the software operation?

A. Hardware Block diagram
B. Board layout
C. Powerpoint
D. Schematic
E. Flowchart

**Question 25** - Which of the following statements accurately describes the intended effect of the declaration `int (*a)[10];`?

A. An array of ten pointers to function.
B. A function that points to an array of ten integers.
C. An array of ten integers.
D. A pointer to an array of ten integers.
E. An array of ten pointers to integers.

**Question 26** - Which of the following should *definitely* be declared using `volatile` keyword?

A. A variable used for a loop.
B. We don't really need volatile keyword, let's take it out in the next standard.
C. A private variable used within a function.
D. A global variable used within an interrupt service routine.
E. A global variable that only used in one function for both reading and writing.

**Question 27** - What is the NZVC flags for the following 16-bit operation?

```
0x1234 - 0x5678

```

A. `N = 0, Z = 0, V = 1, C = 0`
B. `N = 1, Z = 0, V = 0, C = 0`
C. `N = 1, Z = 0, V = 1, C = 1`
D. `N = 1, Z = 0, V = 0, C = 1`
E. `N = 0, Z = 0, V = 0, C = 1`

**Question 28** - Which of the following is responsible for interrupt management and prioritization?

A. Stack
B. RISC
C. Vector Table
D. NVIC
E. SCS

**Question 29** - What is the purpose of Program Counter?

A. Point to the top of the stack.
B. Hold information about program execution, exception number and ALU flags.
C. Point to the next instruction in line.
D. Blocks interrupts.
E. Keep the return address of a subroutine or function call.

**Question 30** - What is the content of the `R0` register after the following code?

```
1  LDR R0, =0x12345678
2  MOV R1, #12
3  LSLS R0, R0, R1
4
```

A. `0x56780000`
B. `0x00045678`
C. `0x45678000`
D. `0x00004567`
E. `0x00001234`

**Question 31** - Which of the following is a state machine that the output depends on the current state and the inputs?

A. NVIC
B. Pipeline
C. FSM
D. Mealy
E. Moore

**Question 32** - How many general purpose register are there on Cortex-M0+?

A. 12
B. 11
C. 16
D. 13
E. 4

**Question 33** - Which of the following can be defined as the specific arrangements of the registers, ALUs, FSMs, memories and other logic building blocks?

A. Microarchitecture
B. Processor
C. Instruction Set Architecture
D. Microcontroller
E. Pipeline

**Question 34** - Which of the following is a memory format that holds the lowest byte in higher address?

A. random access memory
B. read only memory
C. read write memory
D. little endian
E. big endian

**Question 35** - What is the purpose of Link Register?

A. Point to the next instruction in line.
B. Keep the return address of a subroutine or function call.
C. Blocks interrupts.
D. Point to the top of the stack.
E. Hold information about program execution, exception number and ALU flags.

**Question 36** - Which of the following is not a common memory section?

A. text
B. bss
C. data
D. array
E. stack

**Question 37** - Which of the following is a state machine that the output depends only on the current state?

A. Pipeline
B. Moore
C. NVIC
D. FSM
E. Mealy

**Question 38** - Which of the following is a general term for the combination of microprocessor along with RAM, ROM and peripherals (IO) in an integrated Circuit (IC)?

A. Pipeline
B. Microcontroller
C. Microarchitecture
D. Flash
E. Instruction Set Architecture

**Question 39** - What is the result of `R1` register at the end of the following code assuming `mem[0x40001234] = 0x22222222`?

```
 LDR R0, =0x40001234
 LDR R1, [R0]
 LDR R2, =0x0020
 MVN R2, R2
 EORS R1, R2
 STR R1, [R0}

```

A. `0xDDDDDDFD`
B. `0xBFFFEDEB`
C. `0x22222222`
D. `0x40001234`
E. `0x22222202`

**Question 40** - Which of the following is used to describe machines that use simplified instructions that can use many instructions to perform an operations usually completing one instruction per cycle?

A. KISC
B. RISC
C. LISC
D. Harvard
E. FISC